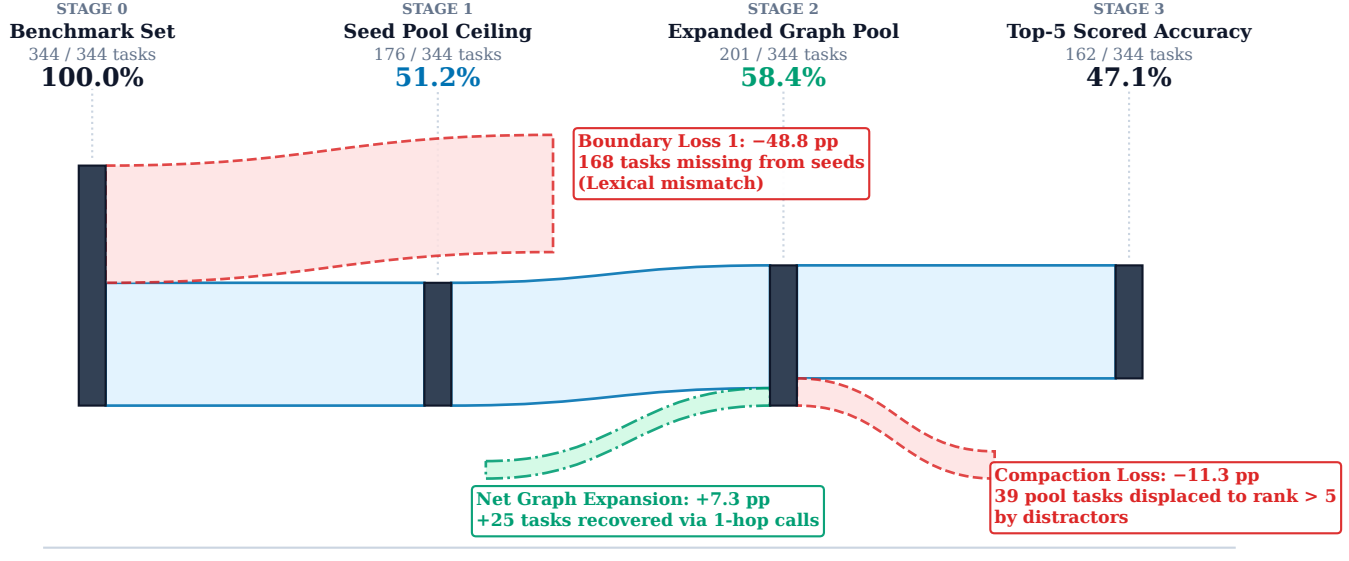

Key Finding: Upstream Boundary Loss (41.6 pp net) exceeds Downstream Compaction Loss (11.3 pp) by **3.68** × (Bottleneck Ratio).